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Assignment 3: Understanding Token Log Probabilities

Objective

.Learn how LLMs assign probabilities to generated tokens

Program

```
from openai import OpenAI
import math

client = OpenAI(api_key=" ")

]
prompt = input("Enter your prompt: ")

response = client.responses.create(
    model="gpt-4.1",
    input=prompt,
    max_output_tokens=100,
    logprobs=True,
)

print("\n=====")
print("Generated Response")
print("=====")
print(response.output_text)

print("\n=====")
print("Token Log Probabilities")
print("=====")

print(f"{'Token':<20} {'Log Prob':<12} {'Probability':<15} {'Cumulative':<15} {'Confidence'}")
print("-" * 90)

cumulative_probability = 1.0

for token_data in response.output[0].content[0].logprobs:
    token = token_data.token
    log_prob = token_data.logprob

    probability = math.exp(log_prob)
```

```
cumulative_probability *= probability
```

```
# Confidence Color
```

```
if probability >= 0.80:
```

```
    confidence = " Green"
```

```
elif probability >= 0.50:
```

```
    confidence = " Yellow"
```

```
else:
```

```
    confidence = " Red"
```

```
print(
```

```
    f"{repr(token):<20} "
```

```
    f"{log_prob:<12.4f} "
```

```
    f"{probability:<15.6f} "
```

```
    f"{cumulative_probability:<15.6f} "
```

```
    f"{confidence}"
```

```
)
```

Output

?Enter your prompt: What is Artificial Intelligence

:Generated Response

.Artificial Intelligence (AI) is the simulation of human intelligence by machines

Token	Log Prob	Probability	Cumulative	Color
Artificial	-0.02	0.9802	0.9802	Green
Intelligence	-0.03	0.9704	0.9512	Green
AI	-0.04	0.9608	0.9047	Green
simulation	-0.80	0.4493	0.2490	Red